Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

INTRODUCTION TO BIOMEDICAL NANOTECHNOLOGY (BME 4053)

Introduction to Biomedical Nanotechnology [BME 4053]

Marks: 50 **Duration: 180 mins.**

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Answer all the questions.

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1a)	A scientist has produced three batches of gold nanoparticles. Propose a simple method to confirm the formation of nanoparticles in these samples, avoiding the use of analytical instruments. Explain the scientific rationale behind your proposed method	(2)
1b)	"As size reduces, surface energy increases." Justify the statement	(3)
1c)	Apply LaMer's theory to explain the process of nanoparticle formation, describing each stage and its significance in achieving controlled particle size.	(5)
2a)	Compare hydrothermal and precipitation methods for nanomaterial synthesis.	(2)
2b)	Propose the critical factors that influence the formation of xerogels during the solgel synthesis with an explanation on sol gel synthesis method.	(3)
2c)	A researcher has a set of reagents: zinc nitrate, sodium hydroxide, cyclohexane, distilled water, and a surfactant. Given the properties of these substances, propose a detailed synthesis strategy for ZnO quantum dots (QDs). Justify your choice of solvents, the role of the surfactant, and the optimal reaction conditions to achieve high-quality QDs.	(5)
3a)	Suggest the modification need to do in the characterization procedure to image the folic acid receptors on a cell using AFM. Justify your suggestion.	(2)
3b)	Suggest a characterization technique to determine the nanomaterial's colloidal stability and explain the instrument's working principle.	(3)
3c)	In an experiment, titanium dioxide nanoparticles were supposed to be attached over a macrophage surface. Suggest an imaging method to confirm the attachment of titanium dioxide nanoparticle over the macrophage surface. Explain the imaging process and principle in detail.	(5)
4a)	Suggest the characterization technique and the procedure to obtain the emission spectra of zinc sulphide doped with manganese quantum dots.	(2)
4b)	The students were synthesizing zinc oxide nanorods functionalized with oleic acid.	(3)

	Suggest a characterization technique to confirm the functionalization of quantum dots with oleic acid. Explain the working principle of the instrument	
4c)	Design and engineer a nanomaterial platform capable of encapsulating and releasing the anticancer drug doxorubicin in a pH-responsive manner.	(5)
5a)	Develop a FET (field effect transistor) based nano biosensor for glucose detection.	(2)
5b)	Design a nanoparticle system for photothermal therapy and deliver a chemotherapeutic drug to the tumor tissues. Justify the design and working principle.	(3)
5c)	Design a nanoparticle contrast agent (a single nanosystem) for optical and MRI multimodal imaging. Justify the design.	(5)

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